Electricity (Set A) — MCQ Worksheet

Science • Chapter 11 • *25 Questions, 1 mark each — Total: 25 marks*

Instructions: *Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.*

Q1. The SI unit of electric current is:

(A) Volt
(B) Ampere
(C) Coulomb
(D) Ohm

Q2. 1 ampere = 1 coulomb / ___

(A) Hour
(B) Minute
(C) Second
(D) Day

Q3. Conventional current flows from:

(A) Negative to positive terminal outside cell
(B) Positive to negative terminal outside cell
(C) No fixed direction
(D) Only inside the cell

Q4. The unit of electric potential difference is:

(A) Ohm
(B) Volt
(C) Watt
(D) Joule

Q5. An ammeter is connected in:

(A) Parallel
(B) Series
(C) Either
(D) None

Q6. A voltmeter is connected in:

(A) Series
(B) Parallel
(C) Both
(D) Open circuit only

Q7. Ohm's law states that V is proportional to I provided:

(A) Resistance is zero
(B) Temperature is constant
(C) Current is zero
(D) Voltage is zero

Q8. The SI unit of resistance is:

(A) Ampere
(B) Volt
(C) Ohm
(D) Coulomb

Q9. The resistance of a wire depends on:

(A) Length only
(B) Area only
(C) Material only
(D) Length, area, material and temperature

Q10. Resistance R of a wire is given by $R = \rho L/A$. Here ρ is the:

(A) Volume
(B) Density
(C) Resistivity
(D) Conductivity

Q11. In series combination of resistors, the equivalent resistance is:

(A) $R_1 + R_2 + R_3$
(B) $1/R_1 + 1/R_2 + 1/R_3$
(C) $R_1 \times R_2 \times R_3$
(D) Always less than R_1

Q12. In parallel combination of resistors, the equivalent resistance:

(A) Equals $R_1 + R_2$
(B) Is less than the smallest resistor
(C) Is greater than the largest
(D) Is zero

Q13. Domestic appliances are connected in parallel because:

(A) Each gets full voltage and works independently
(B) It saves wires
(C) Series wiring is dangerous
(D) Parallel uses less power

Q14. Electrical energy dissipated in a resistor is given by:

(A) $H = IR^2t$
(B) $H = I^2Rt$
(C) $H = IRt$
(D) $H = I^2R/t$

Q15. Power is given by P = ___ ?

(A) VI
(B) V/I
(C) I/V
(D) V + I

Q16. 1 kilowatt-hour (kWh) equals:

(A) 3.6×10^3 J
(B) 3.6×10^5 J
(C) 3.6×10^6 J
(D) 3.6×10^9 J

Q17. A 100 W bulb at 220 V draws a current of approximately:

(A) 0.45 A
(B) 2 A
(C) 4.5 A
(D) 10 A

Q18. The function of an electric fuse is to:

(A) Increase voltage
(B) Reduce voltage
(C) Break the circuit on excess current
(D) Store charge

Q19. Filament of an electric bulb is made of:

(A) Copper
(B) Tungsten
(C) Iron
(D) Aluminium

Q20. The most common reason a fuse melts is:

(A) Overcurrent due to short circuit or overloading
(B) Low voltage
(C) No current
(D) High resistance always

Q21. An earth wire in a 3-pin plug is connected to:

(A) The electric appliance casing/body
(B) The neutral wire
(C) The live wire
(D) Nothing

Q22. In a household, the live wire is colour-coded:

(A) Red
(B) Black
(C) Green
(D) Yellow

Q23. Two resistors of 4 Ω and 6 Ω in series have equivalent resistance:

(A) 2.4 Ω
(B) 10 Ω
(C) 24 Ω
(D) 2 Ω

Q24. Two resistors of 4 Ω and 6 Ω in parallel have equivalent resistance:

(A) 2.4 Ω
(B) 10 Ω
(C) 24 Ω
(D) 12 Ω

Q25. Power dissipated in a resistor of 5 Ω carrying 2 A is:

(A) 10 W
(B) 20 W
(C) 2.5 W
(D) 100 W

— End of Worksheet —